QF620 Additional Examples
Session 2: Binomial Tree and the Risk-Neutral Measure

1 Examples

1. Consider a one-step binomial tree model with $S_0 = 50$, $u = 2$, $r = 1\%$. The actual probability is given by $p = {}^2/_3$, $q = {}^1/_3$.

   (a) Write down the risk-neutral probabilities.

   (b) Evaluate the European call and put option prices struck at $K = 50$ at time $t = 1$.

2. Consider a three-step binomial process with $S_0 = \$4$, $u = 2$, $d = 0.5$, $p = \frac{2}{3}$, $q = \frac{1}{3}$. Use the law of iterated expectation to demonstrate the following relationship:

$$\mathbb{E}_1\left[\mathbb{E}_2[S_3]\right] = \mathbb{E}_1[S_3].$$

3. Consider a three-step binomial tree with parameters: $S_0 = \$4$, $u = 2$, $d = \frac{1}{2}$, $r = 0.25$. Determine the value of

   (a) An European put option struck at $K = \$10$ at time $t = 3$.

   (b) An American put option struck at $K = \$10$ at time $t = 3$.

   (c) An European call option struck at $K = \$10$ at time $t = 3$.

   (d) An American call option struck at $K = \$10$ at time $t = 3$.

4. Consider a binomial tree model where $S_0 = 10$, $u = 2$, $d = {}^1/_2$, $r = 25\%$. The real world probabilities of up and down ticks are given by: $p = {}^3/_4$, $q = {}^1/_4$.

   (a) Explain why should the inequality $0 < d < 1 + r < u$ hold.

   (b) Determine the expectation $\mathbb{E}[S_3]$ under the real world probability measure, and $\mathbb{E}^*[S_3]$ under the risk-neutral probability measure.

   (c) Early exercise premium of an American option is defined as

$$\text{Early Exercise Premium} = V_A - V_E,$$

   where V_A is the American option premium while V_E is the European option premium, both struck at the same level and expire at the same time. Determine the early exercise premium of an American put option struck at $K = 10$ expiring on the 3rd time step.

5. A stock is worth 5 today and the parameters of our binomial tree are $u = 2$, $d = {}^1/_2$, $r = 5\%$. The actual probability is $p = 0.75$, $q = 0.25$.

   (a) What is $\mathbb{E}[S_3]$ under the real-world probability measure?

   (b) Write down the Radon-Nikodym derivative process and use this to show that the discounted stock price is a martingale under the risk-neutral measure.

6. Let S_i denote the stock price at time step i. If the stock price process can be described by the following equation

$$\log\left(\frac{S_{i+1}}{S_i}\right) = \left(r - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\, Z_i,$$

where $i = 0$ to $N-1$, and $Z_i \sim \text{iid}(0,1)$. Derive the expression for S_N.

7. Consider a binomial tree model where $S_0 = 16$, $u = 2$, $d = 1/2$, $r = 25\%$. The real world probabilities of up and down ticks are given by $p = 2/3$, $q = 1/3$. Write down the Radon-Nikodym derivative process $\frac{dQ}{dP}$, and show that

$$\mathbb{E}\left[S_3 \times \frac{dQ}{dP}\right] = \mathbb{E}^*[S_3].$$

8. Consider a Cox-Ross-Rubinstein binomial tree with $S_0 = \$5$, $u = 2$, $r = 4\%$. We would like to value a vanilla option with a strike price of $K = \$10$, maturing on the second time step, i.e. $t = 2$. Calculate the option price if the exercise style and payoff are:

   (a) European put option.

   (b) American put option.

   (c) European call option.

   (d) American call option.

9. **Discussion*** How are u and d in the binomial tree model related to σ, the volatility of the stock? Here, we connect the up and down jump parameters, u and d to the distribution of the underlying asset postulated by our model. To this end, we solve for the u and d parameters that will allow us to match the first and second moments of the underlying asset, respectively. Consider a small time step of Δt, our goal is to ensure that over this time period our binomial tree is consistent with

$$\mathbb{E}[S_{\Delta t}] = p \times u \times S + (1-p) \times d \times S = Se^{r\Delta t}$$
$$\mathbb{E}[S_{\Delta t}^2] = p \times u^2 \times S^2 + (1-p) \times d^2 \times S^2 = S^2 e^{(2r+\sigma^2)\Delta t}.$$

   In Cox-Rubinstein-Ross binomial tree, the relationship between the up and down jump is taken to be $u = \frac{1}{d}$.

2 Detailed Solutions

Throughout, we use the results from the Session 2 notes. With a discretely compounded rate r per step and $0 < d < 1 + r < u$:

$$
\begin{aligned}
p^* &= \frac{S_0(1+r) - S_1^d}{S_1^u - S_1^d} = \frac{(1+r) - d}{u - d}, \qquad q^* = \frac{u - (1+r)}{u - d}, \\
V_n^E &= \frac{1}{1+r}\left[p^* V_{n+1}^u + q^* V_{n+1}^d\right], \\
V_n^A &= \max\left\{\frac{1}{1+r}\left[p^* V_{n+1}^u + q^* V_{n+1}^d\right],\ (K - S_n)^+\right\}, \\
V_0 &= \frac{1}{(1+r)^N}\mathbb{E}^*[V_N] = \frac{1}{(1+r)^N}\sum_{k=0}^{N}\binom{N}{k}(p^*)^{N-k}(q^*)^k\, V_N\left(S_0 u^{N-k} d^k\right), \\
\frac{dQ}{dP}(\omega) &= \frac{Q(\omega)}{P(\omega)}, \qquad \mathbb{E}^Q[X] = \mathbb{E}^P\left[X\,\frac{dQ}{dP}\right], \\
\mathbb{E}_n\big[\mathbb{E}_m[X]\big] &= \mathbb{E}_n[X] \ \ (n \le m), \qquad \frac{S_n}{(1+r)^n} = \mathbb{E}_n^*\left[\frac{S_{n+1}}{(1+r)^{n+1}}\right].
\end{aligned}
$$

For an American call the exercise value $(K - S_n)^+$ is replaced by $(S_n - K)^+$. Since the tosses are independent, each step multiplies the expected price by $(pu + qd)$, so $\mathbb{E}[S_N] = S_0(pu+qd)^N$; under Q, $p^*u + q^*d = 1 + r$ gives $\mathbb{E}^*[S_N] = S_0(1+r)^N$. We use this as a check throughout.

1. (a) The question does not give d, so we take the Cox-Ross-Rubinstein choice $d = 1/u = 0.5$. The stock moves to $S_1^u = 100$ or $S_1^d = 25$, and $0 < 0.5 < 1.01 < 2$, so there is no arbitrage. The risk-neutral probabilities are

$$
\begin{aligned}
p^* &= \frac{(1+r) - d}{u - d} = \frac{1.01 - 0.5}{2 - 0.5} = \frac{0.51}{1.5} = 0.34, \\
q^* &= \frac{u - (1+r)}{u - d} = \frac{2 - 1.01}{2 - 0.5} = \frac{0.99}{1.5} = 0.66.
\end{aligned}
$$

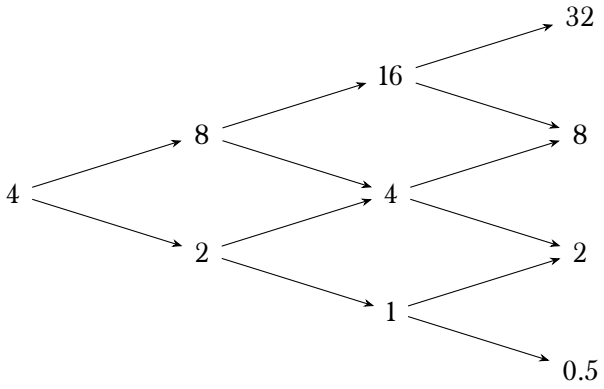
   The real-world probabilities $p = 2/3$, $q = 1/3$ play no role in pricing. ◁

   (b) At $t = 1$ the call pays $(100 - 50)^+ = 50$ in the up state and $(25 - 50)^+ = 0$ in the down state; the put pays 0 and $(50 - 25)^+ = 25$. Discounting the risk-neutral expectation,

$$
\begin{aligned}
C &= \frac{1}{1+r}\left[p^* \times (100 - 50)^+ + q^* \times (25 - 50)^+\right] = \frac{0.34 \times 50 + 0.66 \times 0}{1.01} \\
&= \frac{17}{1.01} = 16.83, \\
P &= \frac{1}{1+r}\left[p^* \times (50 - 100)^+ + q^* \times (50 - 25)^+\right] = \frac{0.34 \times 0 + 0.66 \times 25}{1.01} \\
&= \frac{16.5}{1.01} = 16.34.
\end{aligned}
$$

   Check by put-call parity: $C - P = 0.495$ and $S_0 - K/(1+r) = 50 - 49.505 = 0.495$. ◁

2. First we write down the three-step binomial tree for the stock price ($u = 2$, $d = 0.5$):



Here $p = \frac{2}{3}$, $q = \frac{1}{3}$ are real-world probabilities, and $\mathbb{E}_n$ denotes the expectation conditional on the information at time step n.

Right hand side. From each node at $t = 1$, two further steps give three outcomes with probabilities $p^2 = \frac{4}{9}$, $2pq = \frac{4}{9}$ and $q^2 = \frac{1}{9}$:

$$\mathbb{E}_1[S_3] = \begin{cases} \mathbb{E}_1[S_3 \mid S_1 = 8] = \frac{2}{3} \cdot \frac{2}{3} \cdot 32 + 2 \cdot \frac{2}{3} \cdot \frac{1}{3} \cdot 8 + \frac{1}{3} \cdot \frac{1}{3} \cdot 2 = \frac{128+32+2}{9} = 18, \\ \mathbb{E}_1[S_3 \mid S_1 = 2] = \frac{2}{3} \cdot \frac{2}{3} \cdot 8 + 2 \cdot \frac{2}{3} \cdot \frac{1}{3} \cdot 2 + \frac{1}{3} \cdot \frac{1}{3} \cdot 0.5 = \frac{32+8+0.5}{9} = 4.5. \end{cases}$$

Left hand side. This consists of an inner and an outer expectation:

$$\overbrace{\mathbb{E}_1\Big[\underbrace{\mathbb{E}_2[S_3]}_{\text{inner}}\Big]}^{\text{outer}}.$$

First, we evaluate the inner expectation, one step ahead from each node at $t = 2$:

$$\mathbb{E}_2[S_3] = \begin{cases} \mathbb{E}_2[S_3 \mid S_2 = 16] = \frac{2}{3} \cdot 32 + \frac{1}{3} \cdot 8 = 24, \\ \mathbb{E}_2[S_3 \mid S_2 = 4] = \frac{2}{3} \cdot 8 + \frac{1}{3} \cdot 2 = 6, \\ \mathbb{E}_2[S_3 \mid S_2 = 1] = \frac{2}{3} \cdot 2 + \frac{1}{3} \cdot 0.5 = 1.5. \end{cases}$$

$\mathbb{E}_2[S_3]$ is itself a random variable known at $t = 2$, taking the values 24, 6 and 1.5. Next, we take its expectation conditional on $t = 1$:

$$\mathbb{E}_1\Big[\mathbb{E}_2[S_3]\Big] = \begin{cases} \mathbb{E}_1\Big[\mathbb{E}_2[S_3] \,\Big|\, S_1 = 8\Big] = \frac{2}{3} \cdot 24 + \frac{1}{3} \cdot 6 = 18, \\ \mathbb{E}_1\Big[\mathbb{E}_2[S_3] \,\Big|\, S_1 = 2\Big] = \frac{2}{3} \cdot 6 + \frac{1}{3} \cdot 1.5 = 4.5. \end{cases}$$

Both sides agree at every node at $t = 1$, which verifies $\mathbb{E}_1[\mathbb{E}_2[S_3]] = \mathbb{E}_1[S_3]$ (iterated expectation with $n = 1 \leq m = 2$).

Intuition. Each step multiplies the expected price by $pu + qd = \frac{4}{3} + \frac{1}{6} = 1.5$. Hence $\mathbb{E}_2[S_3] = 1.5\, S_2$ and $\mathbb{E}_1[S_3] = 1.5^2\, S_1 = 2.25\, S_1$, giving 18 and 4.5: averaging one step at a time is the same as averaging two steps at once. ◁

3. With $S_0 = 4$, $u = 2$, $d = 0.5$ and $r = 0.25$, the risk-neutral probabilities and the one-step discount factor are

$$p^* = \frac{1.25 - 0.5}{2 - 0.5} = \frac{1}{2}, \qquad q^* = \frac{1}{2}, \qquad \frac{1}{1+r} = \frac{4}{5}.$$

The stock tree is the same as in Example 2. The terminal stock prices and payoffs are:

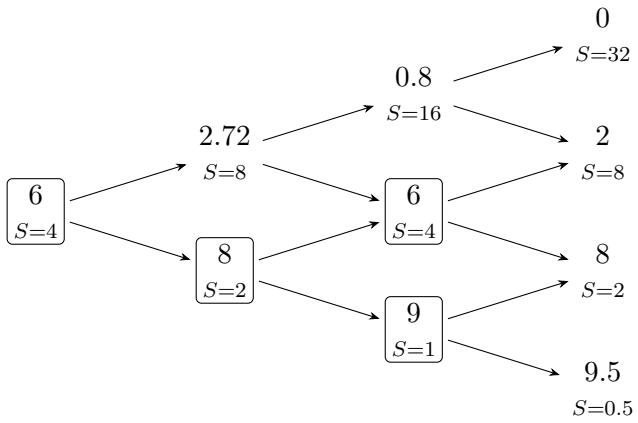
S_3	paths $\binom{3}{k}$	Q-probability	put $(10-S_3)^+$	call $(S_3-10)^+$
32	1	1/8	0	22
8	3	3/8	2	0
2	3	3/8	8	0
0.5	1	1/8	9.5	0

(a) Using the risk-neutral pricing formula,

$$\begin{aligned} V_P^E(0) &= \frac{1}{(1+r)^3}\mathbb{E}^*\big[V_P^E(3)\big] \\ &= \left(\frac{4}{5}\right)^3\left[\frac{1}{2^3}\times 1\times 0+\frac{1}{2^3}\times 3\times 2+\frac{1}{2^3}\times 3\times 8+\frac{1}{2^3}\times 1\times 9.5\right] \\ &= 0.512\times 4.9375 = 2.528. \end{aligned}$$

◁

(b) For the American put we apply backward induction, at each node taking the larger of the continuation value and the exercise value $(10-S_n)^+$. The tree below shows the option value at each node (with the stock price underneath); boxed nodes are where early exercise is optimal.



The backward induction steps are:

$$\begin{aligned} S_2=16: &\quad \max\big\{(10-16)^+,\ \tfrac{4}{5}(\tfrac{1}{2}\cdot 0+\tfrac{1}{2}\cdot 2)\big\} = \max\{0,\ 0.8\} = 0.8 \ \text{(hold)} \\ S_2=4: &\quad \max\big\{(10-4)^+,\ \tfrac{4}{5}(\tfrac{1}{2}\cdot 2+\tfrac{1}{2}\cdot 8)\big\} = \max\{6,\ 4\} = 6 \ \text{(exercise)} \\ S_2=1: &\quad \max\big\{(10-1)^+,\ \tfrac{4}{5}(\tfrac{1}{2}\cdot 8+\tfrac{1}{2}\cdot 9.5)\big\} = \max\{9,\ 7\} = 9 \ \text{(exercise)} \\ S_1=8: &\quad \max\big\{(10-8)^+,\ \tfrac{4}{5}(\tfrac{1}{2}\cdot 0.8+\tfrac{1}{2}\cdot 6)\big\} = \max\{2,\ 2.72\} = 2.72 \ \text{(hold)} \\ S_1=2: &\quad \max\big\{(10-2)^+,\ \tfrac{4}{5}(\tfrac{1}{2}\cdot 6+\tfrac{1}{2}\cdot 9)\big\} = \max\{8,\ 6\} = 8 \ \text{(exercise)} \\ V_P^A: &\quad \max\big\{(10-4)^+,\ \tfrac{4}{5}(\tfrac{1}{2}\cdot 2.72+\tfrac{1}{2}\cdot 8)\big\} = \max\{6,\ 4.288\} = 6 \ \text{(exercise)} \end{aligned}$$

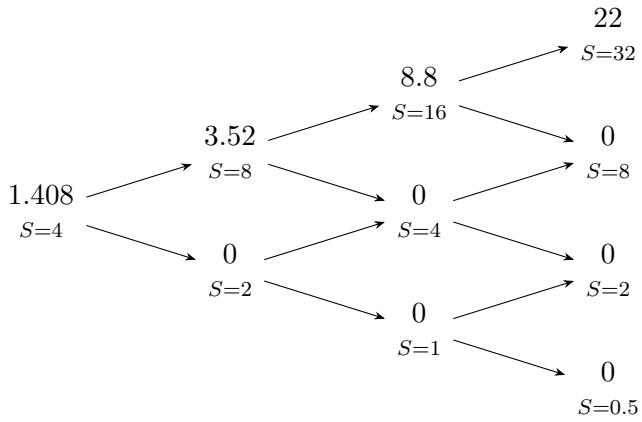
Hence $V_P^A = 6$: exercising today is worth more than waiting. With $r = 25\%$ per step, receiving the strike early is valuable. The early exercise premium is $6 - 2.528 = 3.472$. ◁

(c) Using the risk-neutral pricing formula,

$$\begin{aligned} V_C^E(0) &= \frac{1}{(1+r)^3}\mathbb{E}^*\big[V_C^E(3)\big] \\ &= \left(\frac{4}{5}\right)^3\left[\frac{1}{2^3}\times 1\times 22+\frac{1}{2^3}\times 3\times 0+\frac{1}{2^3}\times 3\times 0+\frac{1}{2^3}\times 1\times 0\right] \\ &= 0.512\times 2.75 = 1.408. \end{aligned}$$

◁

(d) For the American call the exercise value is $(S_n - 10)^+$. Backward induction gives:



$$
\begin{aligned}
S_2 = 16: &\quad \max\left\{(16-10)^+,\ \tfrac{4}{5}(\tfrac{1}{2}\cdot 22 + \tfrac{1}{2}\cdot 0)\right\} = \max\{6,\ 8.8\} = 8.8 \ \text{(hold)} \\
S_2 = 4,\ 1: &\quad \text{both successors and the exercise value are } 0 \ \Rightarrow\ 0 \\
S_1 = 8: &\quad \max\left\{(8-10)^+,\ \tfrac{4}{5}(\tfrac{1}{2}\cdot 8.8 + \tfrac{1}{2}\cdot 0)\right\} = \max\{0,\ 3.52\} = 3.52 \ \text{(hold)} \\
S_1 = 2: &\quad 0 \\
V_C^A: &\quad \max\left\{(4-10)^+,\ \tfrac{4}{5}(\tfrac{1}{2}\cdot 3.52 + \tfrac{1}{2}\cdot 0)\right\} = \max\{0,\ 1.408\} = 1.408 \ \text{(hold)}
\end{aligned}
$$

So $V_C^A = V_C^E = 1.408$. The only node where exercise pays is $S = 16$, and there holding (8.8) beats exercising (6): the continuation value is at least $S_n - K/(1+r) = 16 - 8 = 8 > 6$. As expected, there is no incentive to exercise an American call on a non-dividend-paying stock early, and its early exercise premium is 0.

Check by put-call parity on the European options: $V_C^E - V_P^E = 1.408 - 2.528 = -1.12$ and $S_0 - K/(1+r)^3 = 4 - 10 \times 0.512 = -1.12$. ◁

4. Here $S_0 = 10$, $u = 2$, $d = 0.5$, $r = 0.25$, so $p^* = \frac{1.25-0.5}{2-0.5} = \frac{1}{2}$ and $\frac{1}{1+r} = \frac{4}{5}$. The possible values of S_3 are 80, 20, 5 and 1.25.

   (a) The inequality is the no-arbitrage condition $S_1^d < S_0(1+r) < S_1^u$ divided by S_0:

   - $d > 0$ keeps the stock price strictly positive at every node (limited liability).
   - If $d \geq 1 + r$, borrow S_0 at rate r and buy the stock. At $t = 1$ the stock is worth at least $dS_0 \geq (1+r)S_0$, which repays the loan, with a strictly positive profit in the up state. This is an arbitrage.
   - If $u \leq 1 + r$, short the stock and invest S_0 at rate r. At $t = 1$ we receive $(1+r)S_0$ and owe at most $uS_0 \leq (1+r)S_0$, with a strictly positive profit in the down state. Again an arbitrage.

   Equivalently, $0 < d < 1 + r < u$ is exactly what makes $0 < p^* < 1$, so that Q is a genuine probability measure equivalent to P. Here $0.5 < 1.25 < 2$, so the tree is arbitrage-free. ◁

   (b) Under the real world probability measure ($p = \frac{3}{4}$),

$$
\begin{aligned}
\mathbb{E}[S_3] &= \left(\frac{3}{4}\right)^3 \times 80 + 3 \times \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right) \times 20 \\
&\quad + 3 \times \left(\frac{3}{4}\right) \left(\frac{1}{4}\right)^2 \times 5 + \left(\frac{1}{4}\right)^3 \times 1.25 \\
&= 33.75 + 8.4375 + 0.7031 + 0.0195 = 42.91.
\end{aligned}
$$

   Check: $S_0(pu + qd)^3 = 10 \times (1.5 + 0.125)^3 = 10 \times 1.625^3 = 42.91$.

Under the risk-neutral probability measure ($p^* = \frac{1}{2}$),

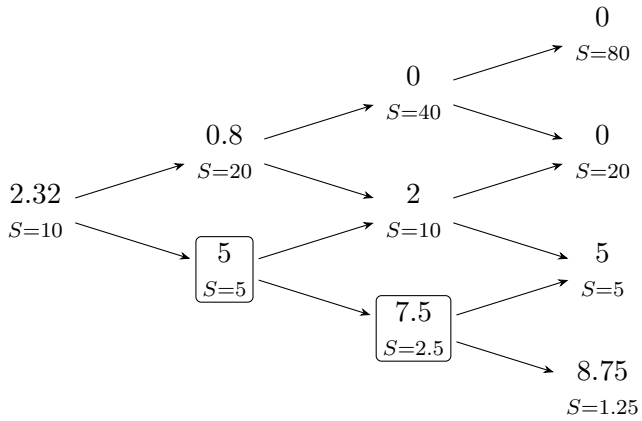
$$\mathbb{E}^*[S_3] = \left(\frac{1}{2}\right)^3 \big[80 + 3\times 20 + 3\times 5 + 1.25\big] = \frac{156.25}{8} = 19.53.$$

Check: $S_0(1+r)^3 = 10 \times 1.953125 = 19.53$, as the martingale property requires. Under P the stock grows by 62.5% per step on average, while under Q it grows at the risk-free 25%. ◁

(c) The put payoffs $(10 - S_3)^+$ at $t = 3$ are 0, 0, 5 and 8.75. The European put is

$$\begin{aligned} V_E &= \left(\frac{4}{5}\right)^3 \left[\frac{1}{8}\times 1\times 0 + \frac{1}{8}\times 3\times 0 + \frac{1}{8}\times 3\times 5 + \frac{1}{8}\times 1\times 8.75\right] \\ &= 0.512 \times 2.96875 = 1.52. \end{aligned}$$

For the American put we use backward induction (boxed nodes: early exercise is optimal):



$$\begin{aligned}
S_2 = 40: &\quad \max\big\{(10-40)^+,\ \tfrac{4}{5}(\tfrac{1}{2}\cdot 0 + \tfrac{1}{2}\cdot 0)\big\} = 0 \\
S_2 = 10: &\quad \max\big\{(10-10)^+,\ \tfrac{4}{5}(\tfrac{1}{2}\cdot 0 + \tfrac{1}{2}\cdot 5)\big\} = \max\{0,\ 2\} = 2 \ \text{(hold)} \\
S_2 = 2.5: &\quad \max\big\{(10-2.5)^+,\ \tfrac{4}{5}(\tfrac{1}{2}\cdot 5 + \tfrac{1}{2}\cdot 8.75)\big\} = \max\{7.5,\ 5.5\} = 7.5 \ \text{(exercise)} \\
S_1 = 20: &\quad \max\big\{(10-20)^+,\ \tfrac{4}{5}(\tfrac{1}{2}\cdot 0 + \tfrac{1}{2}\cdot 2)\big\} = \max\{0,\ 0.8\} = 0.8 \ \text{(hold)} \\
S_1 = 5: &\quad \max\big\{(10-5)^+,\ \tfrac{4}{5}(\tfrac{1}{2}\cdot 2 + \tfrac{1}{2}\cdot 7.5)\big\} = \max\{5,\ 3.8\} = 5 \ \text{(exercise)} \\
V_A: &\quad \max\big\{(10-10)^+,\ \tfrac{4}{5}(\tfrac{1}{2}\cdot 0.8 + \tfrac{1}{2}\cdot 5)\big\} = \max\{0,\ 2.32\} = 2.32 \ \text{(hold)}
\end{aligned}$$

Therefore

$$\text{Early Exercise Premium} = V_A - V_E = 2.32 - 1.52 = 0.80.$$

The premium comes from exercising early at the nodes $S_1 = 5$ and $S_2 = 2.5$. ◁

5. Here $S_0 = 5$, $u = 2$, $d = 0.5$, $r = 0.05$, $p = \frac{3}{4}$, and S_3 takes the values 40, 10, 2.5 and 0.625.

(a) Under the actual probability measure, the expectation of the stock price at time step 3 is

$$\begin{aligned} \mathbb{E}[S_3] &= \left(\frac{3}{4}\right)^3 \times 40 + 3\left(\frac{3}{4}\right)^2\left(\frac{1}{4}\right)\times 10 + 3\left(\frac{3}{4}\right)\left(\frac{1}{4}\right)^2 \times 2.5 + \left(\frac{1}{4}\right)^3 \times 0.625 \\ &= 16.875 + 4.21875 + 0.35156 + 0.00977 = 21.455. \end{aligned}$$

Check: $5 \times 1.625^3 = 21.455$. ◁

(b) The risk-neutral probabilities are

$$p^* = \frac{1.05 - 0.5}{2 - 0.5} = \frac{11}{30} \approx 0.3667, \qquad q^* = \frac{19}{30} \approx 0.6333.$$

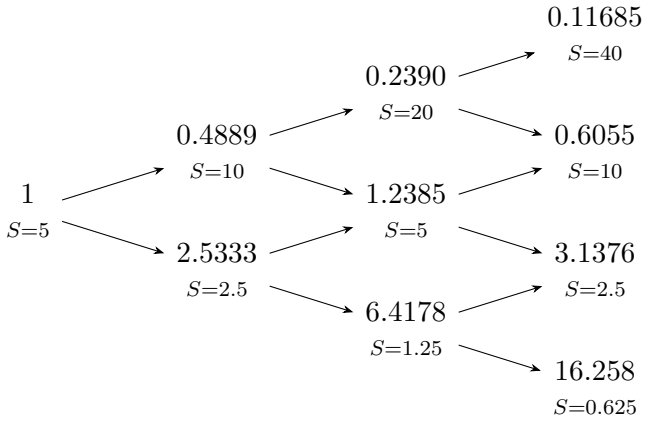
The Radon-Nikodym derivative is the path-by-path ratio $\frac{dQ}{dP}(\omega) = \frac{Q(\omega)}{P(\omega)}$. Because the tosses are independent under both measures, each up move multiplies the ratio by p^*/p and each down move by q^*/q:

$$\frac{p^*}{p} = \frac{11/30}{3/4} = \frac{22}{45} \approx 0.4889, \qquad \frac{q^*}{q} = \frac{19/30}{1/4} = \frac{38}{15} \approx 2.5333.$$

The Radon-Nikodym derivative process $Z_n = \mathbb{E}_n\left[\frac{dQ}{dP}\right]$ after n steps with k down moves is therefore

$$Z_n = \left(\frac{22}{45}\right)^{n-k} \left(\frac{38}{15}\right)^{k},$$

shown below with the stock price under each value:



Z is below 1 after up moves and above 1 after down moves, so Q shifts weight towards low stock prices. Z is also a P-martingale, e.g. $\frac{3}{4}(0.4889) + \frac{1}{4}(2.5333) = 1 = Z_0$.
The expectation of the stock price at time step 3 under the risk-neutral measure is

$$\begin{aligned}\mathbb{E}^*[S_3] &= \left(\frac{11}{30}\right)^3 \times 40 + 3\left(\frac{11}{30}\right)^2\left(\frac{19}{30}\right) \times 10 \\ &\quad + 3\left(\frac{11}{30}\right)\left(\frac{19}{30}\right)^2 \times 2.5 + \left(\frac{19}{30}\right)^3 \times 0.625 \\ &= 5.7881 = 5 \times 1.05^3.\end{aligned}$$

The Radon-Nikodym derivative relates expectations under the two measures as $\mathbb{E}^*[X] = \mathbb{E}[XZ]$. Verifying this,

$$\begin{aligned}\mathbb{E}[S_3 Z_3] &= 0.11685 \times \left(\frac{3}{4}\right)^3 \times 40 + 3 \times 0.6055 \times \left(\frac{3}{4}\right)^2\left(\frac{1}{4}\right) \times 10 \\ &\quad + 3 \times 3.1376 \times \left(\frac{3}{4}\right)\left(\frac{1}{4}\right)^2 \times 2.5 + 16.258 \times \left(\frac{1}{4}\right)^3 \times 0.625 \\ &= 1.9719 + 2.5544 + 1.1031 + 0.1588 = 5.7881 = \mathbb{E}^*[S_3].\end{aligned}$$

The discounted stock price is a martingale under Q. Let $M_n = S_n/(1+r)^n$. For any X that depends on the first $n+1$ tosses, write X^u, X^d for its values after an up or down

move from the current node. Since Z_{n+1}/Z_n equals p^*/p after an up move and q^*/q after a down move,

$$\mathbb{E}_n\left[\frac{Z_{n+1}}{Z_n}X\right] = p\cdot\frac{p^*}{p}\,X^u + q\cdot\frac{q^*}{q}\,X^d = p^*X^u + q^*X^d = \mathbb{E}_n^*[X].$$

Applying this with $X = M_{n+1} = S_{n+1}/1.05^{n+1}$, where $S_{n+1} = uS_n$ or dS_n,

$$\begin{aligned}\mathbb{E}_n^*\left[\frac{S_{n+1}}{1.05^{n+1}}\right] = \mathbb{E}_n\left[\frac{Z_{n+1}}{Z_n}\cdot\frac{S_{n+1}}{1.05^{n+1}}\right] &= \frac{S_n}{1.05^{n+1}}\left[p\cdot\frac{p^*}{p}\cdot u + q\cdot\frac{q^*}{q}\cdot d\right]\\ &= \frac{S_n}{1.05^n}\cdot\frac{p^*u+q^*d}{1.05} = \frac{S_n}{1.05^n},\end{aligned}$$

because $p^*u + q^*d = \frac{11}{30}(2) + \frac{19}{30}(0.5) = \frac{31.5}{30} = 1.05$. This holds at every node, so $M_n = \mathbb{E}_n^*[M_{n+1}]$ and the discounted stock price is a Q-martingale.

As a numerical check, the discounted expectations $\mathbb{E}^*[S_n]/1.05^n = \mathbb{E}[Z_nS_n]/1.05^n$ all equal $S_0 = 5$:

$$\begin{aligned}\text{Step 1:}\quad \frac{\mathbb{E}[Z_1S_1]}{1.05} &= \frac{0.4889\cdot\frac{3}{4}\cdot 10 + 2.5333\cdot\frac{1}{4}\cdot 2.5}{1.05} = \frac{3.6667+1.5833}{1.05} = 5\\ \text{Step 2:}\quad \frac{\mathbb{E}[Z_2S_2]}{1.05^2} &= \frac{1}{1.05^2}\Big[0.2390\cdot\tfrac{9}{16}\cdot 20 + 2\cdot 1.2385\cdot\tfrac{3}{16}\cdot 5\\ &\qquad + 6.4178\cdot\tfrac{1}{16}\cdot 1.25\Big]\\ &= \frac{2.6889+2.3222+0.5014}{1.1025} = \frac{5.5125}{1.1025} = 5\\ \text{Step 3:}\quad \frac{\mathbb{E}[Z_3S_3]}{1.05^3} &= \frac{5.7881}{1.157625} = 5.\end{aligned}$$

On their own these unconditional checks are necessary but not sufficient; the one-step conditional argument above is the proof. ◁

6. We sum both sides from $i = 0$ to $N-1$:

$$\sum_{i=0}^{N-1}\log\left(\frac{S_{i+1}}{S_i}\right) = \sum_{i=0}^{N-1}\left(r-\frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\sum_{i=0}^{N-1}Z_i.$$

Since $\log(S_{i+1}/S_i) = \log S_{i+1} - \log S_i$, the left hand side telescopes:

$$\log\left(\frac{S_1}{S_0}\right) + \log\left(\frac{S_2}{S_1}\right) + \cdots + \log\left(\frac{S_N}{S_{N-1}}\right) = \log S_N - \log S_0.$$

The drift term is a constant added N times. Writing $X = \sum_{i=0}^{N-1} Z_i$ and $T = N\Delta t$,

$$\log S_N - \log S_0 = \left(r-\frac{\sigma^2}{2}\right)N\Delta t + \sigma\sqrt{\Delta t}\,X,$$

$$S_N = S_0\exp\left[\left(r-\frac{\sigma^2}{2}\right)N\Delta t + \sigma\sqrt{\Delta t}\,X\right], \qquad X = \sum_{i=0}^{N-1}Z_i. \qquad \triangleleft$$

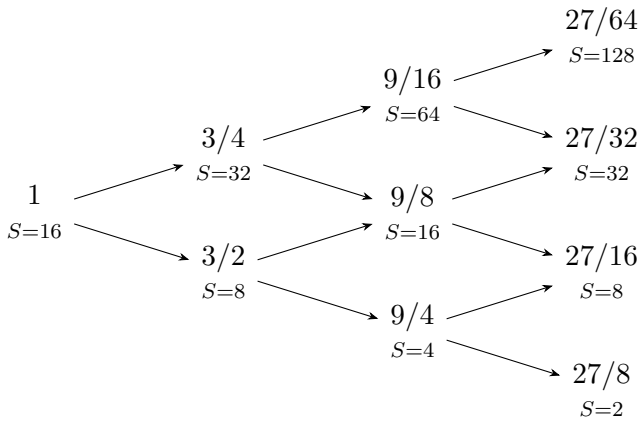
Distribution of X. X is a sum of N independent variables with mean 0 and variance 1, so $\mathbb{E}[X] = 0$ and $V[X] = N$. The noise term $\sigma\sqrt{\Delta t}\,X$ therefore has mean 0 and variance $\sigma^2N\Delta t = \sigma^2T$, matching $\mathbb{E}[\log S_T] = \log S_0 + (r - \frac{\sigma^2}{2})T$ and $V[\log S_T] = \sigma^2T$.

If the Z_i are standard normal, then $X \sim N(0, N)$ exactly, so $X = \sqrt{N}Z$ with $Z \sim N(0,1)$, and since $\sqrt{\Delta t}\sqrt{N} = \sqrt{T}$,

$$S_T = S_0\exp\left[\left(r-\frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}\,Z\right], \qquad Z \sim N(0,1).$$

If the Z_i are only iid with mean 0 and variance 1 (for example ± 1 coin tosses), the Central Limit Theorem gives the same limit as $N \to \infty$. Either way S_T is lognormal. ◁

7. With $S_0 = 16$, $u = 2$, $d = 0.5$, $r = 0.25$ we have $p^* = \frac{1.25-0.5}{2-0.5} = \frac{1}{2}$, $q^* = \frac{1}{2}$. Each up move multiplies $\frac{dQ}{dP}$ by $\frac{p^*}{p} = \frac{1/2}{2/3} = \frac{3}{4}$ and each down move by $\frac{q^*}{q} = \frac{1/2}{1/3} = \frac{3}{2}$. The Radon-Nikodym derivative process (stock price underneath) is:



Under P, the terminal nodes have probabilities $(\frac{2}{3})^3 = \frac{8}{27}$, $3(\frac{2}{3})^2(\frac{1}{3}) = \frac{12}{27}$, $3(\frac{2}{3})(\frac{1}{3})^2 = \frac{6}{27}$ and $(\frac{1}{3})^3 = \frac{1}{27}$. Hence

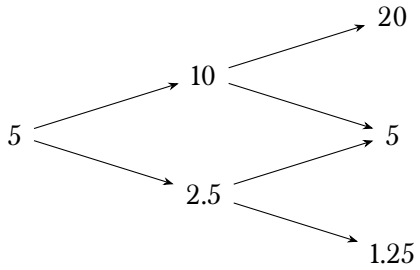
$$\begin{aligned}\mathbb{E}\left[S_3 \cdot \frac{dQ}{dP}\right] &= \left(\frac{2}{3}\right)^3 \cdot 128 \cdot \frac{27}{64} + 3 \cdot \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right) \cdot 32 \cdot \frac{27}{32} \\ &\quad + 3 \cdot \left(\frac{2}{3}\right) \left(\frac{1}{3}\right)^2 \cdot 8 \cdot \frac{27}{16} + \left(\frac{1}{3}\right)^3 \cdot 2 \cdot \frac{27}{8} \\ &= 16 + 12 + 3 + 0.25 = 31.25.\end{aligned}$$

Directly under the risk-neutral measure,

$$\mathbb{E}^*[S_3] = \frac{1}{8} \cdot 128 + \frac{3}{8} \cdot 32 + \frac{3}{8} \cdot 8 + \frac{1}{8} \cdot 2 = 16 + 12 + 3 + 0.25 = 31.25 = 16 \times 1.25^3.$$

The two sums agree term by term because $P(\omega) \cdot \frac{dQ}{dP}(\omega) = Q(\omega)$ for every path; for example $\frac{8}{27} \cdot \frac{27}{64} = \frac{1}{8} = \left(\frac{1}{2}\right)^3$. Hence $\mathbb{E}\left[S_3 \times \frac{dQ}{dP}\right] = \mathbb{E}^*[S_3] = 31.25$. ◁

8. For a Cox-Ross-Rubinstein tree $d = 1/u = 0.5$. The binomial tree is given by:



The risk-neutral probabilities can be calculated as follows:

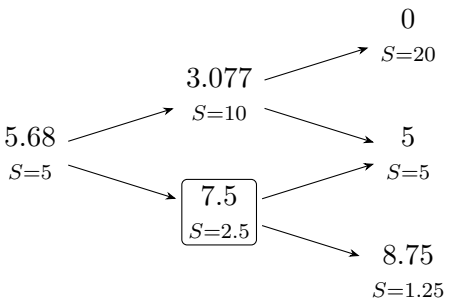
$$p^* = \frac{(1+r)-d}{u-d} = \frac{1.04-0.5}{2-0.5} = 0.36, \qquad q^* = 1 - p^* = 0.64.$$

At $t = 2$ the nodes 20, 5 and 1.25 have Q-probabilities $0.36^2 = 0.1296$, $2(0.36)(0.64) = 0.4608$ and $0.64^2 = 0.4096$.

(a) The European put option can be valued directly as

$$P_0^E = \frac{1}{(1+r)^2}\mathbb{E}^*\big[P_2^E\big] = \frac{1}{1.04^2}\Big[0.36^2\cdot 0 + 2\cdot 0.36\cdot 0.64\cdot 5 + 0.64^2\cdot 8.75\Big]$$
$$= \frac{2.304 + 3.584}{1.0816} \approx 5.44. \qquad \triangleleft$$

(b) For the American put, backward induction with exercise value $(10 - S_n)^+$ gives (boxed node: early exercise is optimal):



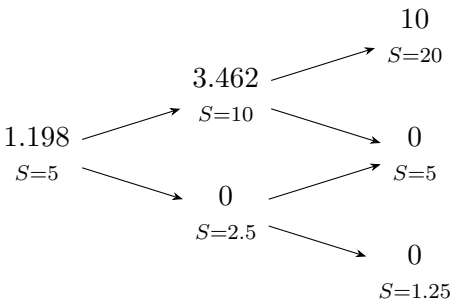
$$S_1 = 10: \quad \max\left\{(10-10)^+,\ \frac{0.36\cdot 0 + 0.64\cdot 5}{1.04}\right\}$$
$$= \max\{0,\ 3.077\} = 3.077 \ \text{(hold)}$$
$$S_1 = 2.5: \quad \max\left\{(10-2.5)^+,\ \frac{0.36\cdot 5 + 0.64\cdot 8.75}{1.04}\right\}$$
$$= \max\{7.5,\ 7.115\} = 7.5 \ \text{(exercise)}$$
$$P_0^A: \quad \max\left\{(10-5)^+,\ \frac{0.36\cdot 3.077 + 0.64\cdot 7.5}{1.04}\right\}$$
$$= \max\{5,\ 5.680\} = 5.68 \ \text{(hold)}$$

The American put is worth 5.68. Its premium over the European put, $5.68 - 5.44 = 0.24$, comes from early exercise at the down node. $\triangleleft$

(c) The European call option can be valued directly as

$$C_0^E = \frac{1}{(1+r)^2}\mathbb{E}^*\big[C_2^E\big] = \frac{1}{1.04^2}\Big[0.36^2\cdot 10 + 2\cdot 0.36\cdot 0.64\cdot 0 + 0.64^2\cdot 0\Big]$$
$$= \frac{1.296}{1.0816} \approx 1.198. \qquad \triangleleft$$

(d) For the American call, backward induction with exercise value $(S_n - 10)^+$ gives:



$$S_1 = 10: \quad \max\left\{(10-10)^+,\ \frac{0.36\cdot 10 + 0.64\cdot 0}{1.04}\right\}$$
$$= \max\{0,\ 3.462\} = 3.462 \ \text{(hold)}$$

$$S_1 = 2.5: \quad \max\left\{(2.5-10)^+,\ 0\right\} = 0$$
$$C_0^A: \quad \max\left\{(5-10)^+,\ \frac{0.36\cdot 3.462 + 0.64\cdot 0}{1.04}\right\}$$
$$= \max\{0,\ 1.198\} = 1.198 \ \text{(hold)}$$

The American call price is 1.198, the same value as the European call: early exercise never pays for a call on a non-dividend-paying stock. Check by put-call parity: $1.198 - 5.444 = -4.246$ and $S_0 - K/(1+r)^2 = 5 - 10/1.0816 = -4.246$. ◁

9. Over a small time step Δt, the tree must reproduce the first two moments of the lognormal model. Compounding is continuous here, so the one-step growth factor $1+r$ becomes $e^{r\Delta t}$:

$$pu + (1-p)d = e^{r\Delta t}, \qquad pu^2 + (1-p)d^2 = e^{(2r+\sigma^2)\Delta t}.$$

The second condition follows from $S_{\Delta t}^2 = S^2 \exp\left[2(r - \frac{\sigma^2}{2})\Delta t + 2\sigma\sqrt{\Delta t}\, Z\right]$ and the normal MGF $\mathbb{E}[e^{2\sigma\sqrt{\Delta t}Z}] = e^{2\sigma^2\Delta t}$, which give $\mathbb{E}[S_{\Delta t}^2] = S^2 e^{(2r+\sigma^2)\Delta t}$.

Step 1: first moment. Solving the first condition for p gives the risk-neutral probability with $1+r$ replaced by $e^{r\Delta t}$:

$$p = \frac{e^{r\Delta t} - d}{u - d}.$$

Step 2: second moment. Write $pu^2 + (1-p)d^2 = p(u^2-d^2) + d^2$ and use $u^2 - d^2 = (u-d)(u+d)$:

$$p(u^2-d^2) + d^2 = (e^{r\Delta t} - d)(u+d) + d^2$$
$$= ue^{r\Delta t} + de^{r\Delta t} - ud - d^2 + d^2.$$

With the Cox-Ross-Rubinstein relation $ud = 1$, the second moment condition becomes

$$ue^{r\Delta t} + de^{r\Delta t} - 1 = e^{(2r+\sigma^2)\Delta t}.$$

Step 3: a quadratic in u. Substitute $d = 1/u$ and multiply through by $ue^{-r\Delta t}$:

$$u^2 + 1 - ue^{-r\Delta t} - ue^{(r+\sigma^2)\Delta t} = 0$$
$$u^2 - u\left(e^{-r\Delta t} + e^{(r+\sigma^2)\Delta t}\right) + 1 = 0.$$

Step 4: solve. Let $A = e^{-r\Delta t} + e^{(r+\sigma^2)\Delta t}$. The product of the two roots is 1, so the roots are exactly u and $d = 1/u$. Taking the larger root for u,

$$u = \frac{e^{-r\Delta t} + e^{(r+\sigma^2)\Delta t}}{2} + \frac{1}{2}\sqrt{\left(e^{-r\Delta t} + e^{(r+\sigma^2)\Delta t}\right)^2 - 4}, \qquad d = \frac{A}{2} - \frac{1}{2}\sqrt{A^2-4}.$$

Step 5: small Δt. Expanding the exponentials to first order around 0, $A \approx (1 - r\Delta t) + (1 + r\Delta t + \sigma^2\Delta t) = 2 + \sigma^2\Delta t$. Then $A^2 - 4 \approx 4\sigma^2\Delta t$ (ignoring $O(\Delta t^2)$), so $\sqrt{A^2-4} \approx 2\sigma\sqrt{\Delta t}$ and

$$u \approx \frac{1}{2}\left(1 - r\Delta t + 1 + r\Delta t + \sigma^2\Delta t\right) + \frac{1}{2}\sqrt{\left(1 - r\Delta t + 1 + r\Delta t + \sigma^2\Delta t\right)^2 - 4}$$
$$= 1 + \sigma\sqrt{\Delta t} + \frac{1}{2}\sigma^2\Delta t + \cdots$$

This matches the Taylor series $e^x = 1 + x + \frac{1}{2}x^2 + \cdots$ with $x = \sigma\sqrt{\Delta t}$, term by term up to order Δt. Hence

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad \text{and by the construction of our model,} \qquad d = \frac{1}{u} = e^{-\sigma\sqrt{\Delta t}}.$$

Consistency check. Each step moves $\log S$ by $\pm\sigma\sqrt{\Delta t}$, and $p \to \frac{1}{2}$ as $\Delta t \to 0$. Each step therefore adds about $\sigma^2\Delta t$ to the variance of $\log S$, and $n = T/\Delta t$ steps give $\sigma^2 T$, as required by the lognormal model. ◁

3 Notes on the Suggested Solutions

Every final number in the suggested solutions is correct. A few labels and statements in the working differ from the solutions above:

- **Example 1**. d is not given in the question; the suggested solution uses $S_1^d = 25$, i.e. the CRR choice $d = 1/u = 0.5$.
- **Example 2**. The tree is described as "two-step" but is the three-step tree; the expectations use the real-world $p = \frac{2}{3}$, not p^*.
- **Example 3(d)**. The American call tree is labelled V_P^A; it should be V_C^A.
- **Example 4(a)**. "No arbitrage" is the right idea; the full answer gives the two arbitrage strategies and the positivity of d.
- **Example 5(b)**. The martingale steps write $\mathbb{E}[S_n]/1.05^n$; this should be $\mathbb{E}[Z_n S_n]/1.05^n = \mathbb{E}^*[S_n]/1.05^n$ (without Z, $\mathbb{E}[S_1]/1.05 = 7.74$). The unconditional check $\mathbb{E}^*[M_n] = M_0$ is necessary but not a proof; the one-step conditional argument is.
- **Example 6**. "$X \sim$ iid$(0, N)$" should read: $X = \sum_i Z_i$ has mean 0 and variance N; it is $N(0, N)$ when the Z_i are normal.